

Gift A. Nwatuze — Curriculum Vitae

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Research Interests

Software Engineering for AI Systems, Artificial Intelligence and Machine Learning, Secure and Trustworthy AI, Privacy-Preserving Machine Learning, Distributed Systems, Cybersecurity.

Profile

Ph.D. candidate in Computer Science with expertise in software engineering, artificial intelligence, and secure distributed systems. Experienced in teaching core undergraduate courses including programming and algorithms, and in developing scalable and secure backend systems. Research focuses on building reliable and privacy-preserving AI systems for real-world applications.

Education

- **Ph.D. in Computer Science** GPA: 3.71/4.00 — *Expected August 2026*
Kent State University, Kent, OH, USA.
- **M.Sc. in Computer Systems Engineering** GPA: 3.89/4.00 — 2019
University of East London, London, UK.
- **B.Sc. in Software Engineering** GPA: 3.62/4.00 — 2018
University of East London, London, UK.

Doctoral Research Experiences

- Federated Learning for Healthcare Secure Data Sharing.
- Quantum-Resilient Federated Learning via Homomorphic-Adaptive Secure Aggregation in Decentralized Environments.
- Zero-Knowledge Proofs for Trustless Data Provenance in Multi-Modal AI Pipelines. Discover Internet of Things.
- Activity of Daily Living (ADL) Recognition using Kinect Data.
- A Framework for Integration of Machine Vision with IoT Sensing.
- Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning.
- DRNMS: An Enhanced Deep Learning-Based System for Data Recovery and Anomaly Detection in Network Monitoring.
- FairAudit: A Causal Explanation Framework for Diagnosing and Mitigating Intersectional Bias.

Teaching and Mentoring Experiences

- **Courses:** CS 13001 (Computer Science 1: Programming and problem solving), assessment design, rubric development, feedback and mentoring, office hours.
- CS 13012 Object Oriented Programming
- CS 13011 Procedural Programming.
- CS-46101/56101-020 Design and Analysis of Algorithms.

Adjunct Instructor — Kent State University

Jan 2026 – Present

- Teach undergraduate and graduate courses in Design and Analysis of Algorithms with emphasis on algorithmic strategies, computational complexity, recursion, graph-based algorithms, dynamic programming, and greedy approaches.
- Design and deliver lectures, assignments, quizzes, examinations, and supplementary instructional materials that support student learning and academic development.
- Assist students in understanding algorithm performance analysis, mathematical reasoning, and practical implementation techniques using programming languages such as Python, C++ and Java.
- Assess and evaluate student performance through assignments, programming projects, presentations, quizzes, and examinations while maintaining university academic standards.
- Hold office hours to provide academic guidance, mentorship, and support on course-related concepts and problem-solving techniques.
- Incorporate modern teaching practices and educational technologies to promote student engagement, participation, and interactive learning experiences.
- Encourage the development of analytical thinking, computational problem-solving abilities, and foundational software engineering skills relevant to real-world computing applications.

Graduate Teaching Assistant — Kent State University

Aug 2022 – Dec.2025

- Delivered instruction in C++ and core programming concepts
- Guided students in algorithm design, recursion, and complexity analysis
- Developed grading rubrics to ensure consistency and fairness
- Provided one-on-one mentoring through office hours and academic support
- Assisted students in building problem-solving and computational thinking skills

Graduate Hourly Employee Coordinator, Dept.of CS. Kent State University Jan 2024 - Dec 2025.

- Oversee and support student employees; ensure task efficiency and compliance with departmental standards.
- Facilitate communication between teams and leadership to improve operations.

Professional Experience

Software Engineering Fellow, Coding Expert Bootcamp (Global Fellowship) Mar 2021 – Sep 2022

- Led design of a bias-aware recommendation engine using explainable AI (XAI) and ethical constraints; results integrated into core curriculum.
- Mentored junior fellows; facilitated workshops on inclusive design and advanced software architecture.

Software Engineer, Trendsville / Pack Media Ltd Jan 2020 – Feb 2021

- Built secure, scalable back-end systems; integrated bias detection and XAI into internal tools; promoted DevOps best practices.

Software Engineer, Tech Nova Solutions May 2019 – Present

- Developed scalable backend systems in distributed environments.
- Designed secure APIs and optimized system performance.
- Implemented CI/CD pipelines and improved system reliability.
- Collaborated across teams to deliver production-level software solutions.

Software Engineer, Civil Service Commission. Jan 2016 – Dec 2019

- Automated public-sector workflows; improved reliability, security, and compliance; mentored junior developers.

Volunteer

Lead Software Architect & Ethical AI Advisor, Techvision Volunteer

Jan 2021 – Dec 2022

- Directed a mobile-first health data platform that increased rural data reliability by ~70%.
- Co-authored a Responsible AI policy framework adopted in emerging-economy contexts; delivered nationwide trainings on ethical AI and digital inclusivity.

Doctoral Research Submission

- Nwatuze, G., Peyravi, H. *Federated Learning for Healthcare Secure Data Sharing* (Submitted to Springer Journal).
- Nwatuze, G., Peyravi, H. *FairAudit: A Causal Explanation Framework for Diagnosing and Mitigating Intersectional Bias* (Submitted to springer ai and ethics journal).
- Nwatuze, G., Peyravi, H. *Zero-Knowledge Proofs for Trustless Data Provenance in Multi-Modal AI Pipelines* (Submitted to ACM Transactions on AI Security and Privacy (TAISAP)).
- Nwatuze, G., Peyravi, H. *Quantum-Resilient Federated Learning via Homomorphic-Adaptive* (Submitted to Springer Journal).

Publications

Doctoral Research Publications

- Nwatuze, Gift, and Hassan Peyravi. "A Framework for Integration of Machine Vision with IoT Sensing." *Sensors* 25.23 (2025): 7237.
- Nwatuze, Gift and Hassan Peyravi. "Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning. In *Proceedings of Computer Science*." *Computer Engineering and Applied Computing*, Cham: Springer Nature Switzerland, 2025.
- Nwatuze, Gift, and Hassan Peyravi. "DRNMS: An Enhanced Deep Learning-Based System for Data Recovery and Anomaly Detection in Network Monitoring." *International Conference on Computational Science and Computational Intelligence*. Cham: Springer Nature Switzerland, 2024.

BSc and MSc Research Publications

- Nwatuze, Gift Aruchi, Lawrence Anebi Enyejo, and Chima Umeaku. "Enhancing cloud data security using a hybrid encryption framework integrating AES, DES, and RC6 with file splitting and steganographic key management." *International Journal of Innovative Science and Research Technology (IJISRT)* 10.1 (2025).
- Nwatuze, Gift Aruchi. "Enhancing Security and Resilience in Cyber-Physical Systems through AI-Driven Threat Detection, Formal Verification, and Blockchain Technologies." *International Journal on Science and Technology (IJSAT)* 3.1 (2025).

- Nwatuze, Gift Aruchi, et al. "Design and evaluation of a user-centric cryptographic model leveraging hybrid algorithms for secure cloud storage and data integrity." American Journal of Innovation in Science and Engineering (AJISE) 4.1 (2025).
- Nwatuze, Gift. "Digital Twin Technology in Software Engineering: Simulation and Testing in Virtual Environments." World Journal of Engineering Research 11 (2025): 322-331.
- Nwatuze, Gift. "Human-Centered Software Engineering: Enhancing Developer Productivity and Well-Being." World Journal of Engineering Research (WJER)11 (2025)
- Nwatuze, Gift. "Blockchain-Based Secure Software Engineering Practices." International Journal for Research in Applied Science and Engineering Technology (IJRASET) 13 (2025).
- Nwatuze, Gift. "Automated Software Refactoring Utilizing Machine Learning for Extensive Systems." International Journal for Research in Applied Science and Engineering Technology (IJRASE) 13 (2025).
- Nwatuze, G. "Adaptive Federated Learning for Secure and Efficient Edge Intelligence in Mobile Computing" World Journal of Engineering Research and Technology (WJERT) 8.0 (2022).
- Nwatuze, G. "Blockchain-Enabled Federated Learning for Trustworthy Multi-Agent Systems" Journal of Engineering Research and Technology (WJERT) 7.0 (2021).
- Nwatuze, G. "Enhanced Intrusion Detection System with Machine Learning Models and Class Imbalance Optimization" Journal of Engineering Research and Technology (WJERT) 5.0 (2019).
- Aruchi, Nwatuze Gift, and Kwok Wai Jing. "Android Device File Encryption and Decryption System." International Journal of Information System and Engineering (IJISE) 7.0 (2019).

Awards & Honors

NSF Travel Award	2025
Certificate of Best Diplomats	2025
Award for Excellence in Innovation	2020
Merit Award — Exceptional Woman in Technology	2019
STEM Outstanding Achievement Award	2018
Dean's Award for Academic Excellence	2015.

Certifications

- CompTIA Security certification
- Cybersecurity Threats and Defense certification
- Cybersecurity operations and controls certification
- Information Security certification

Professional Memberships & Service

IEEE

IEEE Women in Engineering;

ACM

Sigma Xi (Associate)

Nigerian Institution of Professional Engineering and Scientists (NIPES; Fellow).

Conferences & Presentations

- *Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning*, 23rd Int'l Conf. on Scientific Computing (CSC), Las Vegas, USA 2025.
- *Deep learning-based data recovery system for network monitoring (DL-DRSNM)*, Int'l Conf. on Computational Science and Computational Intelligence (CSCI), Las Vegas, USA 2024.
- *Android Device File Encryption and Decryption System*, International Conference Ascent, FMTS College, Cyberjaya Campus, Malaysia 2018.

Peer Review & Project Reviews

Journals: IJIS; FCSIT; IJAIDL; IJAIML, Cogent Arts and Humanities(CAH).

Projects Reviewed (selected): Stock Market Data Analysis; Census Income Patterns; Movie Recommendation; NLP Language Detector; Grocery Sales Forecasting; Asset Pricing ML; Privacy-Preserving House Prices; Heart Disease Prediction; Sentiment Analysis; Customer Churn.